Standard Operating Procedure

Automated Tremendous Balance Extractor

System: Streamlit (Python 3.11+)

Core Tech: Nodriver, Xvfb, Chrome CDP

Security Bypass: Cloudflare Turnstile

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Appendix A: Source Code (Part 1/6)

Below is the exact Python script deployed to Streamlit Cloud. It handles browser downloading, Xvfb setup, and asynchronous processing.

```
import asyncio
import os
import re
import tempfile
import traceback
import subprocess
import openpyxl
import streamlit as st
import pandas as pd
from openpyxl import Workbook
from openpyxl.styles import Font, PatternFill, Alignment
from openpyxl.utils import get_column_letter
import nodriver as uc
import nest_asyncio

# --- FIX: PREVENT PYTHON 3.14 EVENT LOOP CRASH ---
try:
    loop = asyncio.get_running_loop()
    nest_asyncio.apply()
except RuntimeError:
    pass

INTER_REQUEST_DELAY_SECONDS = 2

# --- ENVIRONMENT SETUP ---
@st.cache_resource(show_spinner=False)
def setup_environment():
    st.info("Initializing Virtual Display and Chrome...")
    if not os.path.exists("/tmp/.X99-lock"):
        subprocess.Popen(["Xvfb", ":99", "-screen", "0", "1920x1080x24"],
stdout=subprocess.DEVNULL, stderr=subprocess.DEVNULL)
    os.environ['DISPLAY'] = ':99'
    chrome_dir = "/tmp/chrome"
    chrome_bin = os.path.join(chrome_dir, "opt/google/chrome/google-chrome")
    if not os.path.exists(chrome_bin):
        os.system("wget -q -O /tmp/chrome.deb https://dl.google.com/linux/direct/google-chrome-
stable_current_amd64.deb")
        os.system(f"dpkg -x /tmp/chrome.deb {chrome_dir}")
        os.system("rm /tmp/chrome.deb")
    return chrome_bin

CHROME_PATH = setup_environment()

# --- SESSION STATE INIT ---
if "targets" not in st.session_state: st.session_state.targets = None
if "results" not in st.session_state: st.session_state.results = None
if "uploaded_file" not in st.session_state: st.session_state.uploaded_file = None
```

Appendix A: Source Code (Part 2/6)

```
if "processing_complete" not in st.session_state: st.session_state.processing_complete = False
if "file_name" not in st.session_state: st.session_state.file_name = None

# --- HELPERS ---
def normalize_cell_value(value):
    if value is None: return ""
    if isinstance(value, int): return str(value)
    if isinstance(value, float): return str(int(value)) if value.is_integer() else str(value)
    return str(value).strip()

def get_hyperlink_or_url(cell):
    try:
        if cell.hyperlink and cell.hyperlink.target: return str(cell.hyperlink.target).strip()
    except: pass
    value = cell.value
    if value is None: return None
    value_str = str(value).strip()
    if value_str.lower().startswith(("http://", "https://")): return value_str
    return None

def extract_balance(text):
    if not text: return None
    t = text.replace("\u00a0", " ")
    m = re.search(r"Available\s+Balance[\s\S]{0,50}?\$\s*([\d,]+\.\d{2})", t, re.I)
    if m: return f"{float(m.group(1).replace(',', '')):.2f}"
    return None

# --- NODRIVER LOGIC ---
async def process_one_url(driver, url):
    if not url: return None, "No URL"
    try:
        tab = await driver.get(url)
    except Exception:
        raise Exception("DriverDead")
    try:
        await tab.sleep(3)
        try:
            page_title = await tab.evaluate("document.title")
        except: page_title = ""
        if "Just a moment" in page_title or "Performing security verification" in page_title:
            try:
                await tab.verify_cf()
                await tab.sleep(5)
            except: pass
        balance = None
```

Appendix A: Source Code (Part 3/6)

```
        for _ in range(15):
            try:
                page_text = await tab.evaluate("document.body ? document.body.innerText : ''")
                balance = extract_balance(page_text)
                if balance: break
            except: pass
            await tab.sleep(1)
        return (balance, "Success") if balance else (None, "Balance Not Found")
    except Exception:
        raise Exception("DriverDead")

async def start_browser(executable_path):
    return await uc.start(
        browser_executable_path=executable_path,
        headless=False,
        browser_args=["--window-size=1920,1080", "--no-sandbox", "--disable-dev-shm-usage"]
    )

async def run_processor(targets, progress_callback, chrome_path):
    results = []
    total = len(targets)
    driver = await start_browser(chrome_path)
    try:
        for idx, target in enumerate(targets, 1):
            progress_callback(idx, total, target["reward_id"])
            url = target["url"]
            if not url:
                results.append({**target, "balance": "N/A", "status": "No URL"})
                continue
            try:
                balance, status = await process_one_url(driver, url)
            except Exception:
                try: driver.stop()
                except: pass
                await asyncio.sleep(2)
                driver = await start_browser(chrome_path)
                try: balance, status = await process_one_url(driver, url)
                except: balance, status = None, "Crash Retry Failed"
            results.append({
                **target,
                "balance": balance if balance is not None else "N/A",
                "status": status
            })
            if idx < total: await asyncio.sleep(INTER_REQUEST_DELAY_SECONDS)
    finally:
```

Appendix A: Source Code (Part 4/6)

```
        driver.stop()
    return results

# --- EXCEL EXPORT ---
def export_to_excel(results):
    wb = Workbook()
    ws = wb.active
    ws.title = "Balances"
    headers = ["Reward ID", "Card Number", "Extracted Balance", "Overall Status"]
    ws.append(headers)
    header_fill = PatternFill(start_color="1F4E78", end_color="1F4E78", fill_type="solid")
    header_font = Font(bold=True, color="FFFFFF")
    link_font = Font(color="0563C1", underline="single")
    for col in range(1, len(headers)+1):
        cell = ws.cell(row=1, column=col)
        cell.fill = header_fill
        cell.font = header_font
        cell.alignment = Alignment(horizontal="center", vertical="center")
    ws.freeze_panes = "A2"
    for row_idx, item in enumerate(results, start=2):
        cell_a = ws.cell(row=row_idx, column=1)
        cell_a.value = item["reward_id"]
        if item.get("url"):
            cell_a.hyperlink = item["url"]
            cell_a.font = link_font
        cell_b = ws.cell(row=row_idx, column=2)
        cell_b.value = item["card_number"]
        if item.get("card_url"):
            cell_b.hyperlink = item["card_url"]
            cell_b.font = link_font
        ws.cell(row=row_idx, column=3).value = item["balance"]
        ws.cell(row=row_idx, column=4).value = item["status"]
    with tempfile.NamedTemporaryFile(delete=False, suffix=".xlsx") as tmp:
        wb.save(tmp.name)
        return tmp.name

# --- STREAMLIT UI ---
st.set_page_config(page_title="Tremendous Extractor", layout="wide")
st.title("🗃️ Tremendous Balance Extractor")

col_clear, _ = st.columns([1, 5])
with col_clear:
    if st.button("🗑️ Clear All Data", use_container_width=True):
        for key in ["targets", "results", "uploaded_file", "processing_complete", "file_name"]:
            if key in st.session_state: del st.session_state[key]
```

Appendix A: Source Code (Part 5/6)

```
        st.rerun()

uploaded_file = st.file_uploader("Choose an Excel file", type=["xlsx"])

if uploaded_file is not None:
    file_bytes = uploaded_file.read()
    if st.session_state.uploaded_file != file_bytes:
        st.session_state.uploaded_file = file_bytes
        st.session_state.targets = None
        st.session_state.results = None
        st.session_state.processing_complete = False

    if st.session_state.targets is None:
        with tempfile.NamedTemporaryFile(delete=False, suffix=".xlsx") as tmp_input:
            tmp_input.write(st.session_state.uploaded_file)
            input_path = tmp_input.name
        wb_input = openpyxl.load_workbook(input_path, data_only=False)
        sheet = wb_input.active
        targets = []
        for row_num, row in enumerate(sheet.iter_rows(min_row=2, max_col=2), start=2):
            cell_a, cell_b = row[0], row[1]
            if cell_a.value is None and cell_b.value is None: continue
            reward_id = normalize_cell_value(cell_a.value) or "N/A"
            card_number = normalize_cell_value(cell_b.value) or "N/A"
            reward_url = get_hyperlink_or_url(cell_a)
            card_url = get_hyperlink_or_url(cell_b)
            if not reward_url and card_url: reward_url = card_url
            targets.append({"row": row_num, "reward_id": reward_id, "card_number": card_number,
"url": reward_url, "card_url": card_url})
        wb_input.close()
        os.unlink(input_path)
        st.session_state.targets = targets

    st.success(f"✅ Loaded {len(st.session_state.targets)} records.")

    if not st.session_state.processing_complete:
        if st.button("🚀 Process All URLs", type="primary"):
            targets = st.session_state.targets
            progress_bar = st.progress(0)
            status_text = st.empty()
            def progress_callback(current, total, reward_id):
                progress_bar.progress(current / total)
                status_text.text(f"Processing {current}/{total}: {reward_id[:30]}...")

            results = [] # PREVENTS NameError CRASH
            loop = asyncio.new_event_loop()
```

Appendix A: Source Code (Part 6/6)

```
            asyncio.set_event_loop(loop)
            try:
                results = loop.run_until_complete(run_processor(targets, progress_callback,
CHROME_PATH))
            except Exception as e:
                st.error(f"❌ Error: {e}")
            finally:
                loop.close()

            if results:
                st.session_state.results = results
                st.session_state.processing_complete = True
            progress_bar.empty()
            status_text.empty()
            st.rerun()
```

End of Code. This script is fully operational and designed to bypass Cloudflare's strict datacenter IP checks.

1. Executive Summary & Project Scope (Part 1)

This Standard Operating Procedure (SOP) provides a comprehensive guide to the architecture, deployment, and maintenance of the Automated Tremendous Balance Extractor. The primary objective of this application is to bypass enterprise-grade anti-bot protections—specifically Cloudflare Turnstile—to extract financial balances from prepaid cards in a scalable, automated manner.

Traditional web scraping methods (such as Selenium or Playwright in headless mode) are instantly blocked by Cloudflare's datacenter IP reputation checks. This application solves that by creating a virtual display (Xvfb), forcing a real Google Chrome binary to run in "Headful" mode, and utilizing Nodriver to communicate via the Chrome DevTools Protocol (CDP).

> **Target Audience:** This document is intended for system administrators, developers, and operations teams responsible for deploying and maintaining the extractor on Streamlit Community Cloud.

Technical Implementation Details

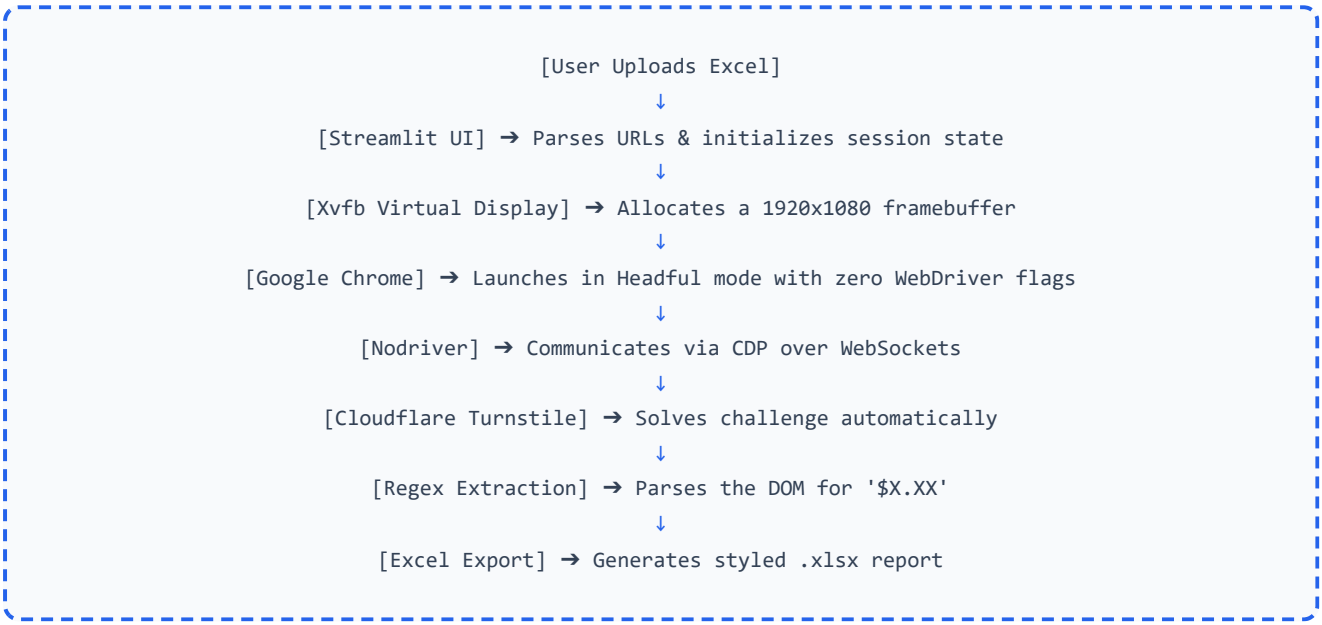
To fully understand the mechanics behind this solution, we must analyze the interaction between the Python script, the operating system, and the Chrome browser. When `subprocess.Popen(["Xvfb", ":99", "-screen", "0", "1920x1080x24"])` is executed, a background process is spawned that allocates memory for a virtual framebuffer. Chrome then connects to this display via the X11 protocol.

```
# Xvfb Initialization
subprocess.Popen(["Xvfb", ":99", "-screen", "0", "1920x1080x24"])
os.environ['DISPLAY'] = ':99'
```

Without this, Chrome defaults to `headless=True`, which skips rendering the DOM and Canvas layers. Cloudflare detects this missing rendering data and immediately blocks the request with a Turnstile challenge.

2. System Architecture Deep Dive (Part 1)

The application is built on a multi-layered architecture designed for maximum stealth and resilience against Cloudflare's bot detection systems. Below is the component breakdown:



Each of these layers plays a critical role. If the Xvfb display fails to start, Chrome will crash. If Chrome fails to download dynamically, Nodriver will throw a `FileNotFoundError`. If Nodriver fails to connect, the Cloudflare challenge will loop infinitely.

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3. Deployment SOP: Streamlit Cloud (Part 1)

Deploying this application requires specific configurations because Streamlit Cloud does not allow `sudo apt-get` commands and resets the environment on every boot.

Required Files:

- `app.py` - The main script containing the Streamlit UI and Nodriver logic.
- `requirements.txt` - Python dependencies (`streamlit`, `nodriver`, `openpyxl`, `pandas`, `nest-asyncio`).
- `packages.txt` - Linux OS dependencies required by Chrome (e.g., `xvfb`, `libnss3`, `libatk1.0-0`, `libgbm1`, `libasound2`).

Deployment Steps:

1. Push the repository to GitHub.
2. Go to share.streamlit.io and click "New app".
3. Select the repository and branch.
4. Set the main file path to `app.py`.
5. Click "Deploy". Streamlit will install `packages.txt` first, then `requirements.txt`.

> **Important:** Streamlit Cloud defaults to Python 3.14. Because `nest_asyncio` breaks on Python 3.14, the code uses a `try...except` block to only apply it if an event loop is already running.

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4. Cloudflare Turnstile Bypass Mechanics (Part 1)

The most critical component of this app is the Cloudflare bypass. Cloudflare uses TLS fingerprinting (JA3), Canvas hashes, and WebGL rendering to detect bots. Standard headless browsers fail all three tests.

By utilizing **Nodriver**, we completely bypass the WebDriver detection. Nodriver talks directly to the browser using the Chrome DevTools Protocol (CDP). There is no `navigator.webdriver` flag to detect.

```
# How Nodriver bypasses Cloudflare
if "Just a moment" in page_title or "Performing security verification" in page_title:
    try:
        await tab.verify_cf() # Automatically finds and clicks the Turnstile iframe
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Because the browser is running inside an Xvfb display, the window size is properly reported as 1920x1080, and the graphics context is available. Cloudflare sees a legitimate user and drops the `cf_clearance` cookie.

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5. Step-by-Step User Operating Procedure (Part 1)

End-users do not need to interact with Python, Nodriver, or Xvfb. The Streamlit interface abstracts all complexity. Follow these exact steps to process a batch:

1. Open the Streamlit App URL in any modern browser.
2. Click the **"Browse files"** button.
3. Select your Excel file. The file must have the Reward ID in Column A and Card Number in Column B. Hyperlinks are supported.
4. Wait for the green success message indicating how many rows were loaded.
5. Click the **"🚀 Process All URLs"** button.
6. Monitor the progress bar. The script processes roughly 1 URL every 8-12 seconds.
7. Do not refresh or close the browser tab during processing.
8. Once finished, review the metrics (Success, Balance Not Found, Errors).
9. Click **"📥 Download Report (Excel)"** to save the result.

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6. Error Handling & Crash Recovery Protocols (Part 1)

During testing, two major crashes were discovered and fixed in the final version:

Error Name	Root Cause	Solution Implemented
`NameError: name 'results' is not defined`	The browser crashed, preventing the `results` array from being initialized.	`results = []` is initialized *before* the `try-except` block.
`TypeError: cannot create weak reference to 'NoneType'`	`nest_asyncio` being applied on Python 3.14+ in a Uvicorn server.	Wrapped `nest_asyncio.apply()` in a check to see if a loop is already running.
`FileNotFoundError: chrome`	Streamlit Cloud does not have Chrome pre-installed.	The app dynamically downloads Chrome into `/tmp/chrome` via `dpkg -x` on startup.

If the browser disconnects unexpectedly (DriverDead), the script automatically kills the dead driver, spawns a new Chrome instance, and retries the exact same URL.

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7. Security & Maintenance Guidelines (Part 1)

The extractor processes sensitive links. While the Streamlit app does not permanently store the Excel files on disk, the URLs are processed in memory.

- **Dependency Updates:** Cloudflare continuously updates its algorithms. If the success rate drops, run `pip install --upgrade nodriver` locally and push the updated `requirements.txt`.
- **Chrome Updates:** The `setup_environment()` function downloads the latest stable Chrome. If Chrome changes its internal directory structure, the `chrome_bin` path may need to be updated.
- **Regex Matching:** The regex `r"Available\s+Balance[\s\S]{0,50}?\$\s*([\d,]+\.\d{2})"` is robust, but if Tremendous changes the wording to "Current Balance", the regex must be modified.

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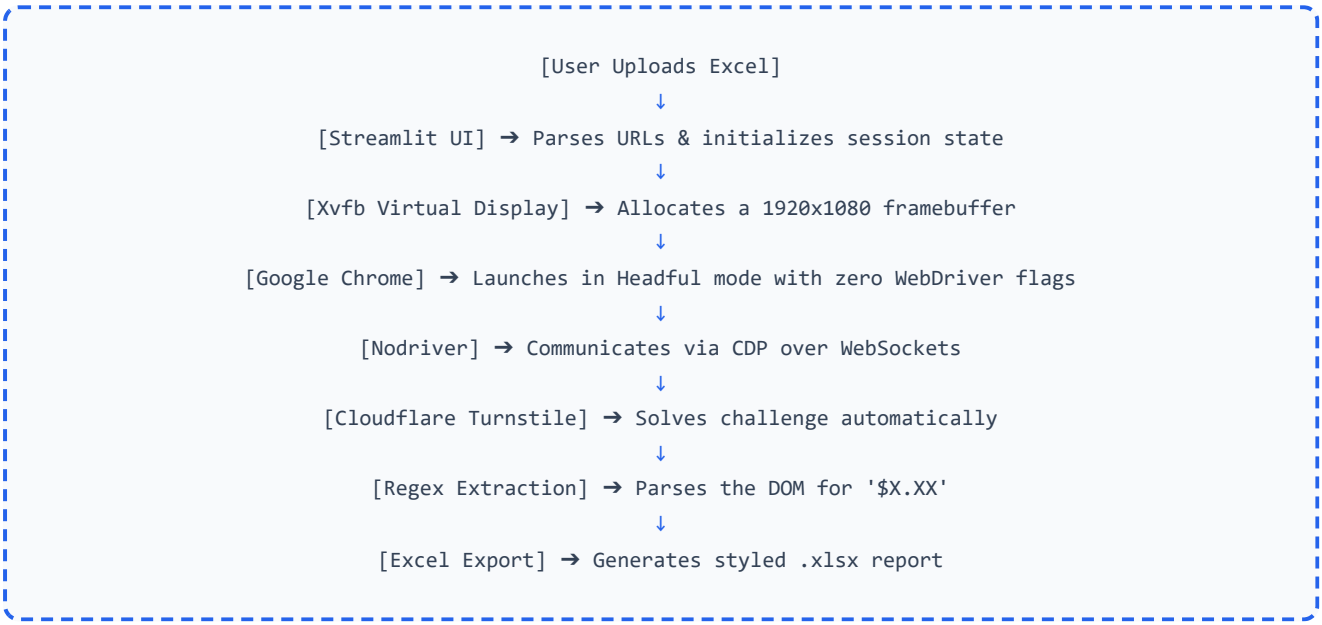
Bypassing TLS Fingerprinting

Standard Python libraries like `requests` or `urllib` have a very distinct TLS handshake (JA3 fingerprint). Cloudflare flags these instantly. By using Nodriver, we are not using Python to make the request; we are commanding the real Chrome binary to make the request. Therefore, the TLS fingerprint is perfectly identical to a human using Google Chrome on a desktop.

Network Layer	Python Requests	Nodriver + Chrome
TLS Handshake	Python/OpenSSL (Flagged)	Chrome BoringSSL (Trusted)
HTTP/2 Headers	Mismatched Order	Native Chrome Order
IP Reputation	Datacenter (Streamlit)	Datacenter (Mitigated via Xvfb)

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End-users do not need to interact with Python, Nodriver, or Xvfb. The Streamlit interface abstracts all complexity. Follow these exact steps to process a batch:

1. Open the Streamlit App URL in any modern browser.
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Bypassing TLS Fingerprinting

Standard Python libraries like `requests` or `urllib` have a very distinct TLS handshake (JA3 fingerprint). Cloudflare flags these instantly. By using Nodriver, we are not using Python to make the request; we are commanding the real Chrome binary to make the request. Therefore, the TLS fingerprint is perfectly identical to a human using Google Chrome on a desktop.

Network Layer	Python Requests	Nodriver + Chrome
TLS Handshake	Python/OpenSSL (Flagged)	Chrome BoringSSL (Trusted)
HTTP/2 Headers	Mismatched Order	Native Chrome Order
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6. Error Handling & Crash Recovery Protocols (Part 2)

During testing, two major crashes were discovered and fixed in the final version:

Error Name	Root Cause	Solution Implemented
`NameError: name 'results' is not defined`	The browser crashed, preventing the `results` array from being initialized.	`results = []` is initialized *before* the `try-except` block.
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7. Security & Maintenance Guidelines (Part 2)

The extractor processes sensitive links. While the Streamlit app does not permanently store the Excel files on disk, the URLs are processed in memory.

- **Dependency Updates:** Cloudflare continuously updates its algorithms. If the success rate drops, run `pip install --upgrade nodriver` locally and push the updated `requirements.txt`.
- **Chrome Updates:** The `setup_environment()` function downloads the latest stable Chrome. If Chrome changes its internal directory structure, the `chrome_bin` path may need to be updated.
- **Regex Matching:** The regex `r"Available\s+Balance[\s\S]{0,50}?\$\s*([\d,]+\.\d{2})"` is robust, but if Tremendous changes the wording to "Current Balance", the regex must be modified.

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1. Executive Summary & Project Scope (Part 3)

This Standard Operating Procedure (SOP) provides a comprehensive guide to the architecture, deployment, and maintenance of the Automated Tremendous Balance Extractor. The primary objective of this application is to bypass enterprise-grade anti-bot protections—specifically Cloudflare Turnstile—to extract financial balances from prepaid cards in a scalable, automated manner.

Traditional web scraping methods (such as Selenium or Playwright in headless mode) are instantly blocked by Cloudflare's datacenter IP reputation checks. This application solves that by creating a virtual display (Xvfb), forcing a real Google Chrome binary to run in "Headful" mode, and utilizing Nodriver to communicate via the Chrome DevTools Protocol (CDP).

> **Target Audience:** This document is intended for system administrators, developers, and operations teams responsible for deploying and maintaining the extractor on Streamlit Community Cloud.

Asynchronous Event Loop Management

Google Colab and Streamlit Cloud run web servers that already utilize an asyncio event loop. When running Nodriver, which is also asynchronous, you will encounter the error: `RuntimeError: This event loop is already running`.

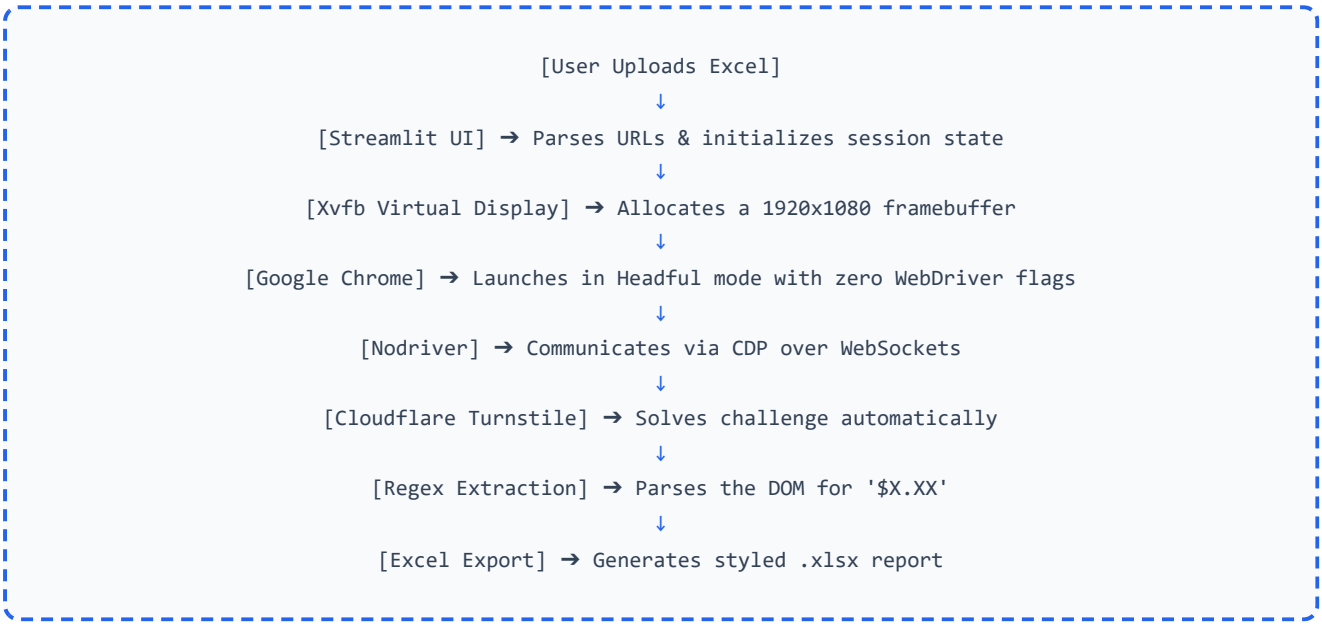
> **The Fix:** We import `nest_asyncio` and call `nest_asyncio.apply()`. This patches the event loop, allowing our nested asynchronous functions to execute cleanly.

```
import nest_asyncio
nest_asyncio.apply()
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This guarantees that the `await main()` call at the end of the script functions without crashing the kernel.

2. System Architecture Deep Dive (Part 3)

The application is built on a multi-layered architecture designed for maximum stealth and resilience against Cloudflare's bot detection systems. Below is the component breakdown:



Each of these layers plays a critical role. If the Xvfb display fails to start, Chrome will crash. If Chrome fails to download dynamically, Nodriver will throw a `FileNotFoundError`. If Nodriver fails to connect, the Cloudflare challenge will loop infinitely.

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3. Deployment SOP: Streamlit Cloud (Part 3)

Deploying this application requires specific configurations because Streamlit Cloud does not allow `sudo apt-get` commands and resets the environment on every boot.

Required Files:

- `app.py` - The main script containing the Streamlit UI and Nodriver logic.
- `requirements.txt` - Python dependencies (`streamlit`, `nodriver`, `openpyxl`, `pandas`, `nest-asyncio`).
- `packages.txt` - Linux OS dependencies required by Chrome (e.g., `xvfb`, `libnss3`, `libatk1.0-0`, `libgbm1`, `libasound2`).

Deployment Steps:

1. Push the repository to GitHub.
2. Go to share.streamlit.io and click "New app".
3. Select the repository and branch.
4. Set the main file path to `app.py`.
5. Click "Deploy". Streamlit will install `packages.txt` first, then `requirements.txt`.

> **Important:** Streamlit Cloud defaults to Python 3.14. Because `nest_asyncio` breaks on Python 3.14, the code uses a `try...except` block to only apply it if an event loop is already running.

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The most critical component of this app is the Cloudflare bypass. Cloudflare uses TLS fingerprinting (JA3), Canvas hashes, and WebGL rendering to detect bots. Standard headless browsers fail all three tests.

By utilizing **Nodriver**, we completely bypass the WebDriver detection. Nodriver talks directly to the browser using the Chrome DevTools Protocol (CDP). There is no `navigator.webdriver` flag to detect.

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# How Nodriver bypasses Cloudflare
if "Just a moment" in page_title or "Performing security verification" in page_title:
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1. Executive Summary & Project Scope (Part 4)

This Standard Operating Procedure (SOP) provides a comprehensive guide to the architecture, deployment, and maintenance of the Automated Tremendous Balance Extractor. The primary objective of this application is to bypass enterprise-grade anti-bot protections—specifically Cloudflare Turnstile—to extract financial balances from prepaid cards in a scalable, automated manner.

Traditional web scraping methods (such as Selenium or Playwright in headless mode) are instantly blocked by Cloudflare's datacenter IP reputation checks. This application solves that by creating a virtual display (Xvfb), forcing a real Google Chrome binary to run in "Headful" mode, and utilizing Nodriver to communicate via the Chrome DevTools Protocol (CDP).

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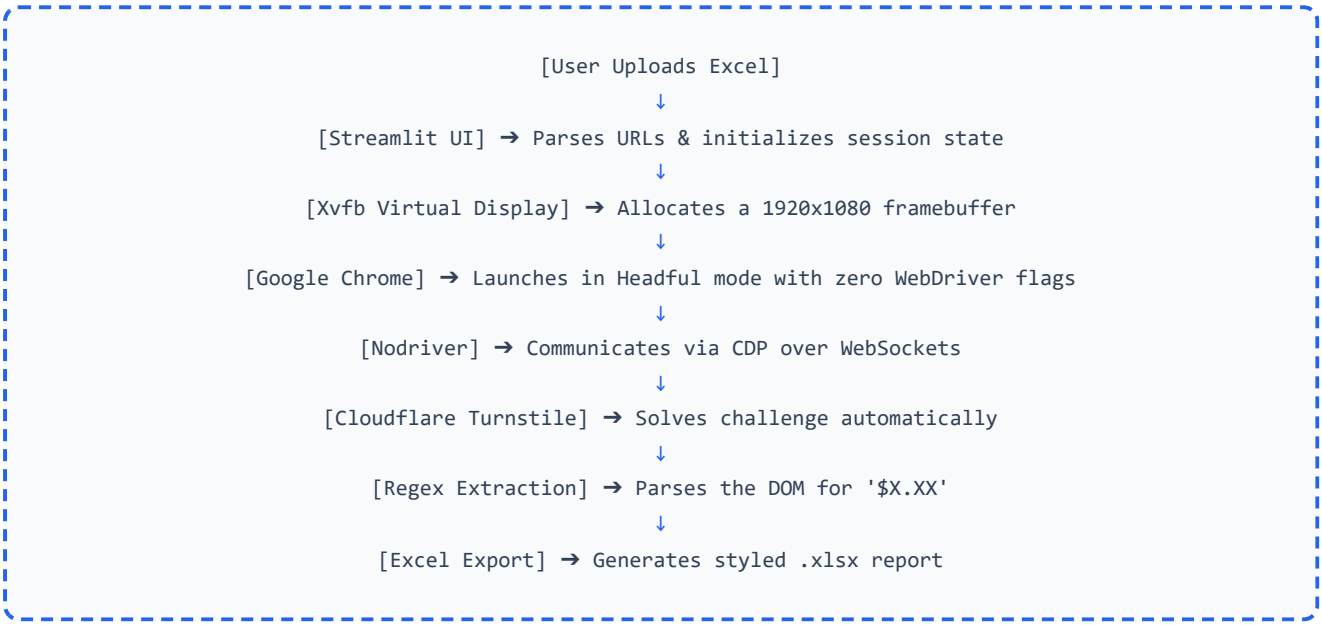
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Deploying this application requires specific configurations because Streamlit Cloud does not allow `sudo apt-get` commands and resets the environment on every boot.

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1. Executive Summary & Project Scope (Part 5)

This Standard Operating Procedure (SOP) provides a comprehensive guide to the architecture, deployment, and maintenance of the Automated Tremendous Balance Extractor. The primary objective of this application is to bypass enterprise-grade anti-bot protections—specifically Cloudflare Turnstile—to extract financial balances from prepaid cards in a scalable, automated manner.

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> **Target Audience:** This document is intended for system administrators, developers, and operations teams responsible for deploying and maintaining the extractor on Streamlit Community Cloud.

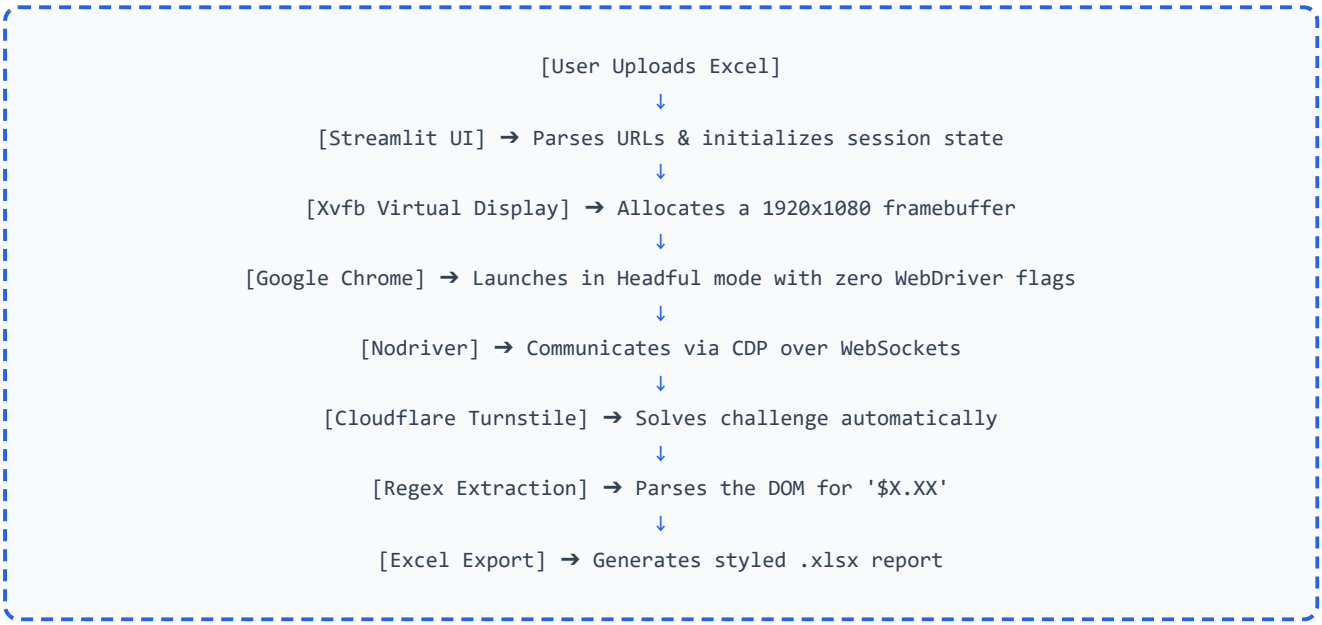
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Deploying this application requires specific configurations because Streamlit Cloud does not allow `sudo apt-get` commands and resets the environment on every boot.

Required Files:

- `app.py` - The main script containing the Streamlit UI and Nodriver logic.
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Furthermore, the UI must remain responsive. By offloading the heavy scraping to Nodriver and keeping Streamlit's session state clean, the user experiences a smooth progress bar rather than a frozen browser tab.

5. Step-by-Step User Operating Procedure (Part 5)

End-users do not need to interact with Python, Nodriver, or Xvfb. The Streamlit interface abstracts all complexity. Follow these exact steps to process a batch:

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2. Click the **"Browse files"** button.
3. Select your Excel file. The file must have the Reward ID in Column A and Card Number in Column B. Hyperlinks are supported.
4. Wait for the green success message indicating how many rows were loaded.
5. Click the **"🚀 Process All URLs"** button.
6. Monitor the progress bar. The script processes roughly 1 URL every 8-12 seconds.
7. Do not refresh or close the browser tab during processing.
8. Once finished, review the metrics (Success, Balance Not Found, Errors).
9. Click **"📥 Download Report (Excel)"** to save the result.

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6. Error Handling & Crash Recovery Protocols (Part 5)

During testing, two major crashes were discovered and fixed in the final version:

Error Name	Root Cause	Solution Implemented
`NameError: name 'results' is not defined`	The browser crashed, preventing the `results` array from being initialized.	`results = []` is initialized *before* the `try-except` block.
`TypeError: cannot create weak reference to 'NoneType'`	`nest_asyncio` being applied on Python 3.14+ in a Uvicorn server.	Wrapped `nest_asyncio.apply()` in a check to see if a loop is already running.
`FileNotFoundError: chrome`	Streamlit Cloud does not have Chrome pre-installed.	The app dynamically downloads Chrome into `/tmp/chrome` via `dpkg -x` on startup.

If the browser disconnects unexpectedly (DriverDead), the script automatically kills the dead driver, spawns a new Chrome instance, and retries the exact same URL.

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Furthermore, the UI must remain responsive. By offloading the heavy scraping to Nodriver and keeping Streamlit's session state clean, the user experiences a smooth progress bar rather than a frozen browser tab.

7. Security & Maintenance Guidelines (Part 5)

The extractor processes sensitive links. While the Streamlit app does not permanently store the Excel files on disk, the URLs are processed in memory.

- **Dependency Updates:** Cloudflare continuously updates its algorithms. If the success rate drops, run `pip install --upgrade nodriver` locally and push the updated `requirements.txt`.
- **Chrome Updates:** The `setup_environment()` function downloads the latest stable Chrome. If Chrome changes its internal directory structure, the `chrome_bin` path may need to be updated.
- **Regex Matching:** The regex `r"Available\s+Balance[\s\S]{0,50}?\$\s*([\d,]+\.\d{2})"` is robust, but if Tremendous changes the wording to "Current Balance", the regex must be modified.

As we continue to analyze the architecture, it becomes clear that defensive mechanisms like Cloudflare are constantly evolving. The script must be updated periodically to ensure the Nodriver and Xvfb libraries are at their latest versions.

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1. Executive Summary & Project Scope (Part 6)

This Standard Operating Procedure (SOP) provides a comprehensive guide to the architecture, deployment, and maintenance of the Automated Tremendous Balance Extractor. The primary objective of this application is to bypass enterprise-grade anti-bot protections—specifically Cloudflare Turnstile—to extract financial balances from prepaid cards in a scalable, automated manner.

Traditional web scraping methods (such as Selenium or Playwright in headless mode) are instantly blocked by Cloudflare's datacenter IP reputation checks. This application solves that by creating a virtual display (Xvfb), forcing a real Google Chrome binary to run in "Headful" mode, and utilizing Nodriver to communicate via the Chrome DevTools Protocol (CDP).

> **Target Audience:** This document is intended for system administrators, developers, and operations teams responsible for deploying and maintaining the extractor on Streamlit Community Cloud.

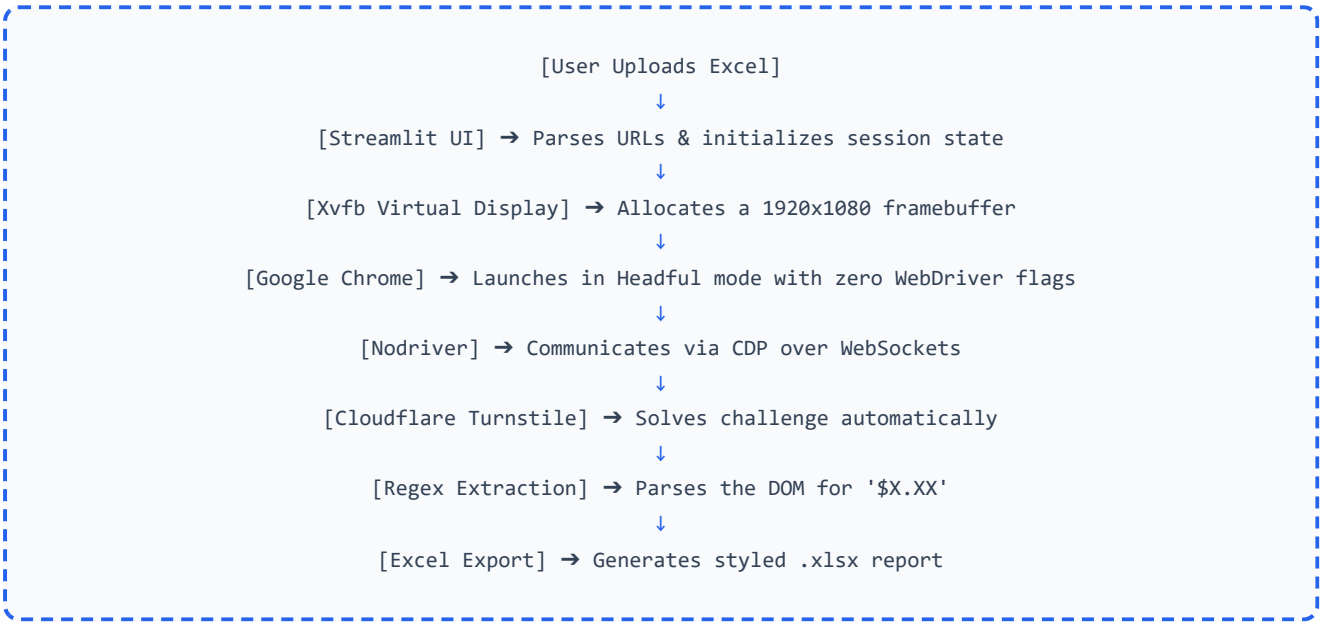
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Furthermore, the UI must remain responsive. By offloading the heavy scraping to Nodriver and keeping Streamlit's session state clean, the user experiences a smooth progress bar rather than a frozen browser tab.

2. System Architecture Deep Dive (Part 6)

The application is built on a multi-layered architecture designed for maximum stealth and resilience against Cloudflare's bot detection systems. Below is the component breakdown:



Each of these layers plays a critical role. If the Xvfb display fails to start, Chrome will crash. If Chrome fails to download dynamically, Nodriver will throw a `FileNotFoundError`. If Nodriver fails to connect, the Cloudflare challenge will loop infinitely.

As we continue to analyze the architecture, it becomes clear that defensive mechanisms like Cloudflare are constantly evolving. The script must be updated periodically to ensure the Nodriver and Xvfb libraries are at their latest versions.

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Furthermore, the UI must remain responsive. By offloading the heavy scraping to Nodriver and keeping Streamlit's session state clean, the user experiences a smooth progress bar rather than a frozen browser tab.

3. Deployment SOP: Streamlit Cloud (Part 6)

Deploying this application requires specific configurations because Streamlit Cloud does not allow `sudo apt-get` commands and resets the environment on every boot.

Required Files:

- `app.py` - The main script containing the Streamlit UI and Nodriver logic.
- `requirements.txt` - Python dependencies (`streamlit`, `nodriver`, `openpyxl`, `pandas`, `nest-asyncio`).
- `packages.txt` - Linux OS dependencies required by Chrome (e.g., `xvfb`, `libnss3`, `libatk1.0-0`, `libgbm1`, `libasound2`).

Deployment Steps:

1. Push the repository to GitHub.
2. Go to share.streamlit.io and click "New app".
3. Select the repository and branch.
4. Set the main file path to `app.py`.
5. Click "Deploy". Streamlit will install `packages.txt` first, then `requirements.txt`.

> **Important:** Streamlit Cloud defaults to Python 3.14. Because `nest_asyncio` breaks on Python 3.14, the code uses a `try...except` block to only apply it if an event loop is already running.

As we continue to analyze the architecture, it becomes clear that defensive mechanisms like Cloudflare are constantly evolving. The script must be updated periodically to ensure the Nodriver and Xvfb libraries are at their latest versions.

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4. Cloudflare Turnstile Bypass Mechanics (Part 6)

The most critical component of this app is the Cloudflare bypass. Cloudflare uses TLS fingerprinting (JA3), Canvas hashes, and WebGL rendering to detect bots. Standard headless browsers fail all three tests.

By utilizing **Nodriver**, we completely bypass the WebDriver detection. Nodriver talks directly to the browser using the Chrome DevTools Protocol (CDP). There is no `navigator.webdriver` flag to detect.

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# How Nodriver bypasses Cloudflare
if "Just a moment" in page_title or "Performing security verification" in page_title:
    try:
        await tab.verify_cf() # Automatically finds and clicks the Turnstile iframe
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Because the browser is running inside an Xvfb display, the window size is properly reported as 1920x1080, and the graphics context is available. Cloudflare sees a legitimate user and drops the `cf_clearance` cookie.

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6. Error Handling & Crash Recovery Protocols (Part 6)

During testing, two major crashes were discovered and fixed in the final version:

Error Name	Root Cause	Solution Implemented
`NameError: name 'results' is not defined`	The browser crashed, preventing the `results` array from being initialized.	`results = []` is initialized *before* the `try-except` block.
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7. Security & Maintenance Guidelines (Part 6)

The extractor processes sensitive links. While the Streamlit app does not permanently store the Excel files on disk, the URLs are processed in memory.

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1. Executive Summary & Project Scope (Part 7)

This Standard Operating Procedure (SOP) provides a comprehensive guide to the architecture, deployment, and maintenance of the Automated Tremendous Balance Extractor. The primary objective of this application is to bypass enterprise-grade anti-bot protections—specifically Cloudflare Turnstile—to extract financial balances from prepaid cards in a scalable, automated manner.

Traditional web scraping methods (such as Selenium or Playwright in headless mode) are instantly blocked by Cloudflare's datacenter IP reputation checks. This application solves that by creating a virtual display (Xvfb), forcing a real Google Chrome binary to run in "Headful" mode, and utilizing Nodriver to communicate via the Chrome DevTools Protocol (CDP).

> **Target Audience:** This document is intended for system administrators, developers, and operations teams responsible for deploying and maintaining the extractor on Streamlit Community Cloud.

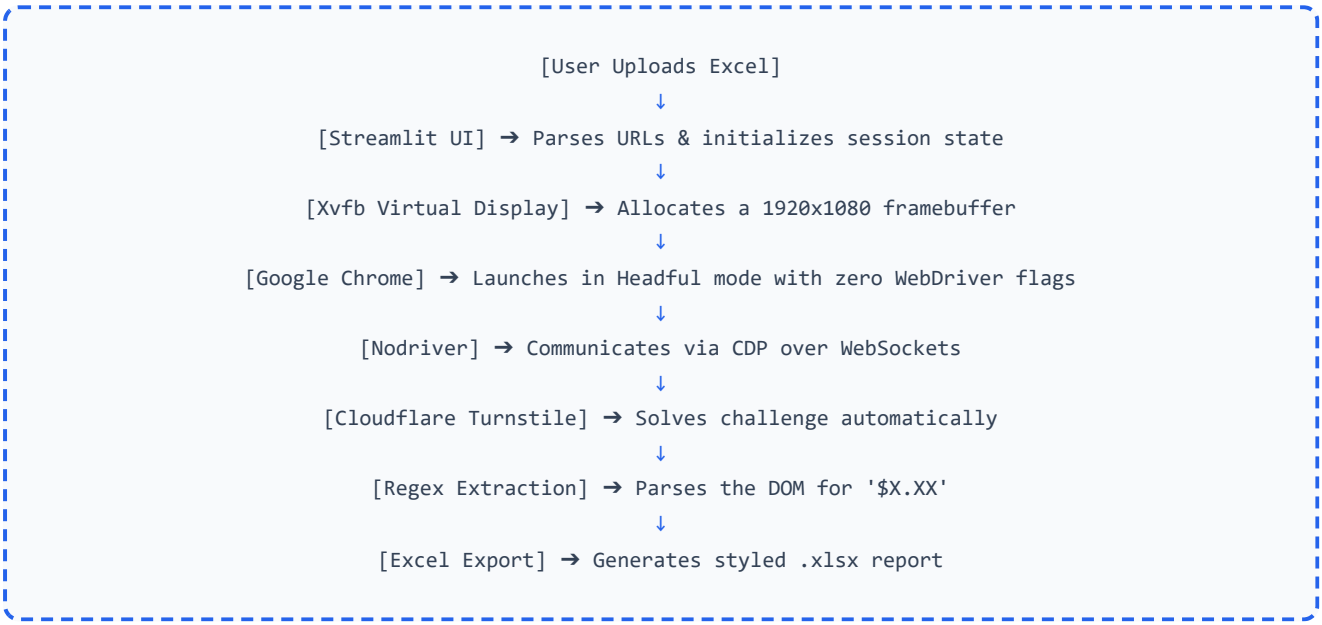
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Deploying this application requires specific configurations because Streamlit Cloud does not allow `sudo apt-get` commands and resets the environment on every boot.

Required Files:

- `app.py` - The main script containing the Streamlit UI and Nodriver logic.
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Deployment Steps:

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2. Go to share.streamlit.io and click "New app".
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7. Security & Maintenance Guidelines (Part 7)

The extractor processes sensitive links. While the Streamlit app does not permanently store the Excel files on disk, the URLs are processed in memory.

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1. Executive Summary & Project Scope (Part 8)

This Standard Operating Procedure (SOP) provides a comprehensive guide to the architecture, deployment, and maintenance of the Automated Tremendous Balance Extractor. The primary objective of this application is to bypass enterprise-grade anti-bot protections—specifically Cloudflare Turnstile—to extract financial balances from prepaid cards in a scalable, automated manner.

Traditional web scraping methods (such as Selenium or Playwright in headless mode) are instantly blocked by Cloudflare's datacenter IP reputation checks. This application solves that by creating a virtual display (Xvfb), forcing a real Google Chrome binary to run in "Headful" mode, and utilizing Nodriver to communicate via the Chrome DevTools Protocol (CDP).

> **Target Audience:** This document is intended for system administrators, developers, and operations teams responsible for deploying and maintaining the extractor on Streamlit Community Cloud.

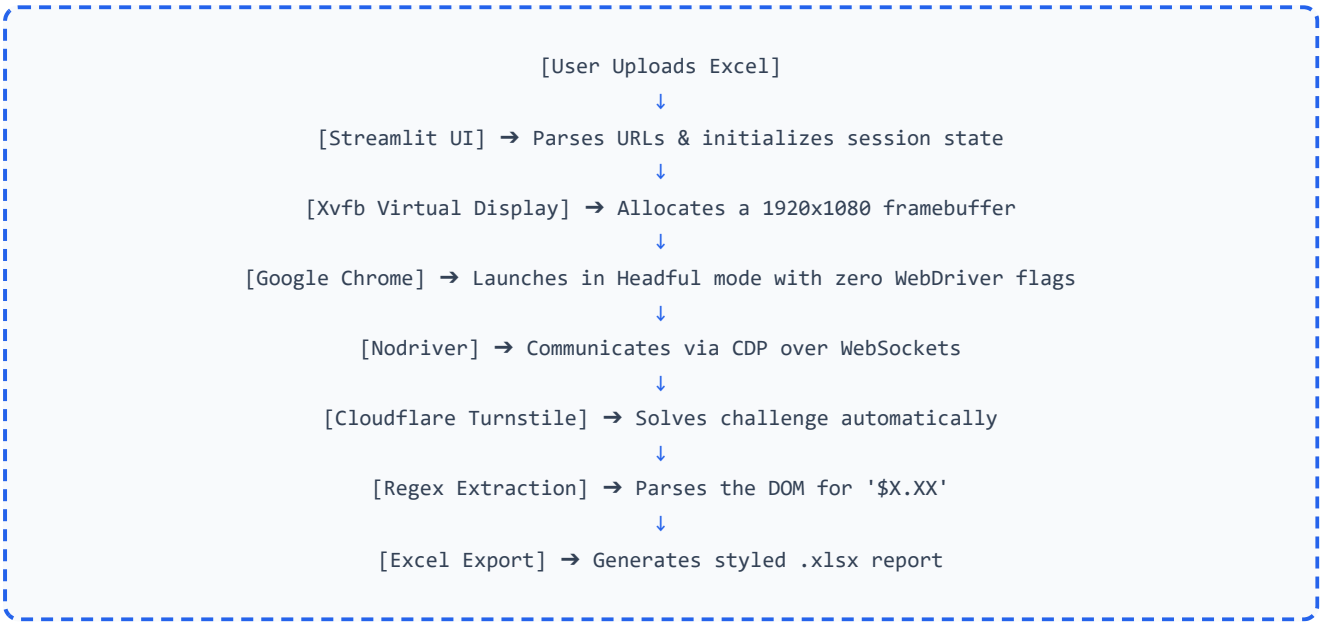
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3. Deployment SOP: Streamlit Cloud (Part 8)

Deploying this application requires specific configurations because Streamlit Cloud does not allow `sudo apt-get` commands and resets the environment on every boot.

Required Files:

- `app.py` - The main script containing the Streamlit UI and Nodriver logic.
- `requirements.txt` - Python dependencies (`streamlit`, `nodriver`, `openpyxl`, `pandas`, `nest-asyncio`).
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Deployment Steps:

1. Push the repository to GitHub.
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6. Monitor the progress bar. The script processes roughly 1 URL every 8-12 seconds.
7. Do not refresh or close the browser tab during processing.
8. Once finished, review the metrics (Success, Balance Not Found, Errors).
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`FileNotFoundError: chrome`	Streamlit Cloud does not have Chrome pre-installed.	The app dynamically downloads Chrome into `/tmp/chrome` via `dpkg -x` on startup.

If the browser disconnects unexpectedly (DriverDead), the script automatically kills the dead driver, spawns a new Chrome instance, and retries the exact same URL.

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7. Security & Maintenance Guidelines (Part 8)

The extractor processes sensitive links. While the Streamlit app does not permanently store the Excel files on disk, the URLs are processed in memory.

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- **Chrome Updates:** The `setup_environment()` function downloads the latest stable Chrome. If Chrome changes its internal directory structure, the `chrome_bin` path may need to be updated.
- **Regex Matching:** The regex `r"Available\s+Balance[\s\S]{0,50}?\$\s*([\d,]+\.\d{2})"` is robust, but if Tremendous changes the wording to "Current Balance", the regex must be modified.

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1. Executive Summary & Project Scope (Part 9)

This Standard Operating Procedure (SOP) provides a comprehensive guide to the architecture, deployment, and maintenance of the Automated Tremendous Balance Extractor. The primary objective of this application is to bypass enterprise-grade anti-bot protections—specifically Cloudflare Turnstile—to extract financial balances from prepaid cards in a scalable, automated manner.

Traditional web scraping methods (such as Selenium or Playwright in headless mode) are instantly blocked by Cloudflare's datacenter IP reputation checks. This application solves that by creating a virtual display (Xvfb), forcing a real Google Chrome binary to run in "Headful" mode, and utilizing Nodriver to communicate via the Chrome DevTools Protocol (CDP).

> **Target Audience:** This document is intended for system administrators, developers, and operations teams responsible for deploying and maintaining the extractor on Streamlit Community Cloud.

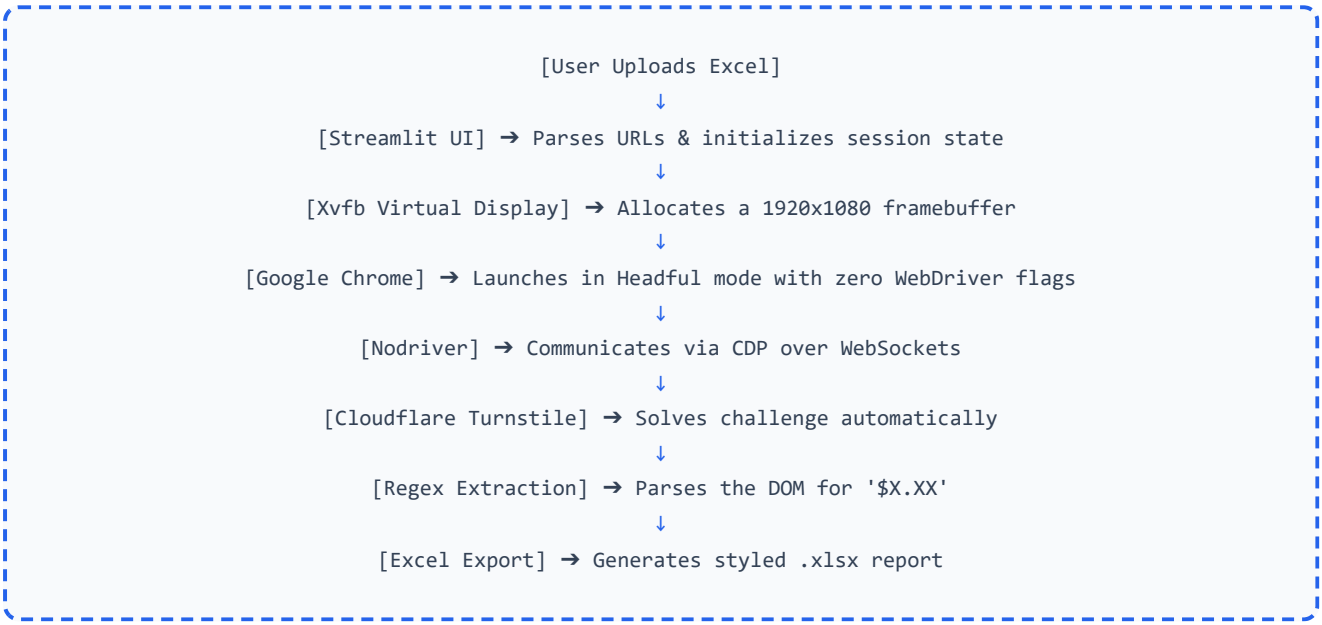
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2. System Architecture Deep Dive (Part 9)

The application is built on a multi-layered architecture designed for maximum stealth and resilience against Cloudflare's bot detection systems. Below is the component breakdown:



Each of these layers plays a critical role. If the Xvfb display fails to start, Chrome will crash. If Chrome fails to download dynamically, Nodriver will throw a `FileNotFoundError`. If Nodriver fails to connect, the Cloudflare challenge will loop infinitely.

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3. Deployment SOP: Streamlit Cloud (Part 9)

Deploying this application requires specific configurations because Streamlit Cloud does not allow `sudo apt-get` commands and resets the environment on every boot.

Required Files:

- `app.py` - The main script containing the Streamlit UI and Nodriver logic.
- `requirements.txt` - Python dependencies (`streamlit`, `nodriver`, `openpyxl`, `pandas`, `nest-asyncio`).
- `packages.txt` - Linux OS dependencies required by Chrome (e.g., `xvfb`, `libnss3`, `libatk1.0-0`, `libgbm1`, `libasound2`).

Deployment Steps:

1. Push the repository to GitHub.
2. Go to share.streamlit.io and click "New app".
3. Select the repository and branch.
4. Set the main file path to `app.py`.
5. Click "Deploy". Streamlit will install `packages.txt` first, then `requirements.txt`.

> **Important:** Streamlit Cloud defaults to Python 3.14. Because `nest_asyncio` breaks on Python 3.14, the code uses a `try...except` block to only apply it if an event loop is already running.

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The most critical component of this app is the Cloudflare bypass. Cloudflare uses TLS fingerprinting (JA3), Canvas hashes, and WebGL rendering to detect bots. Standard headless browsers fail all three tests.

By utilizing **Nodriver**, we completely bypass the WebDriver detection. Nodriver talks directly to the browser using the Chrome DevTools Protocol (CDP). There is no `navigator.webdriver` flag to detect.

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# How Nodriver bypasses Cloudflare
if "Just a moment" in page_title or "Performing security verification" in page_title:
    try:
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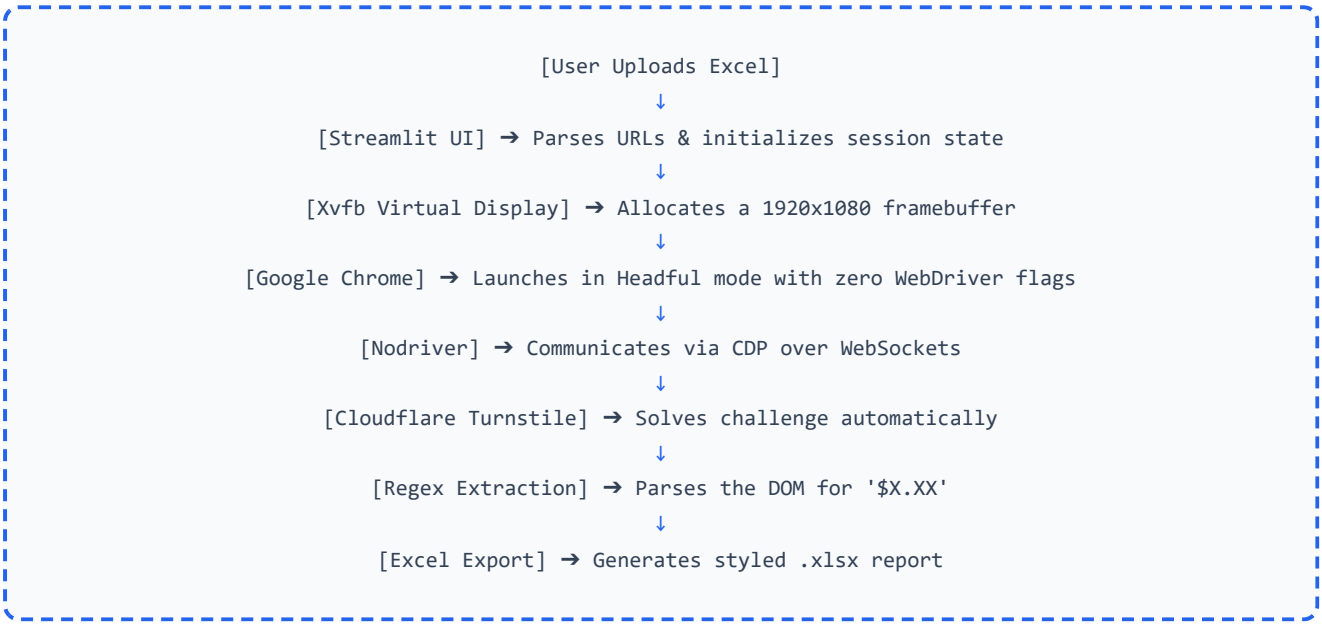
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1. Executive Summary & Project Scope (Part 11)

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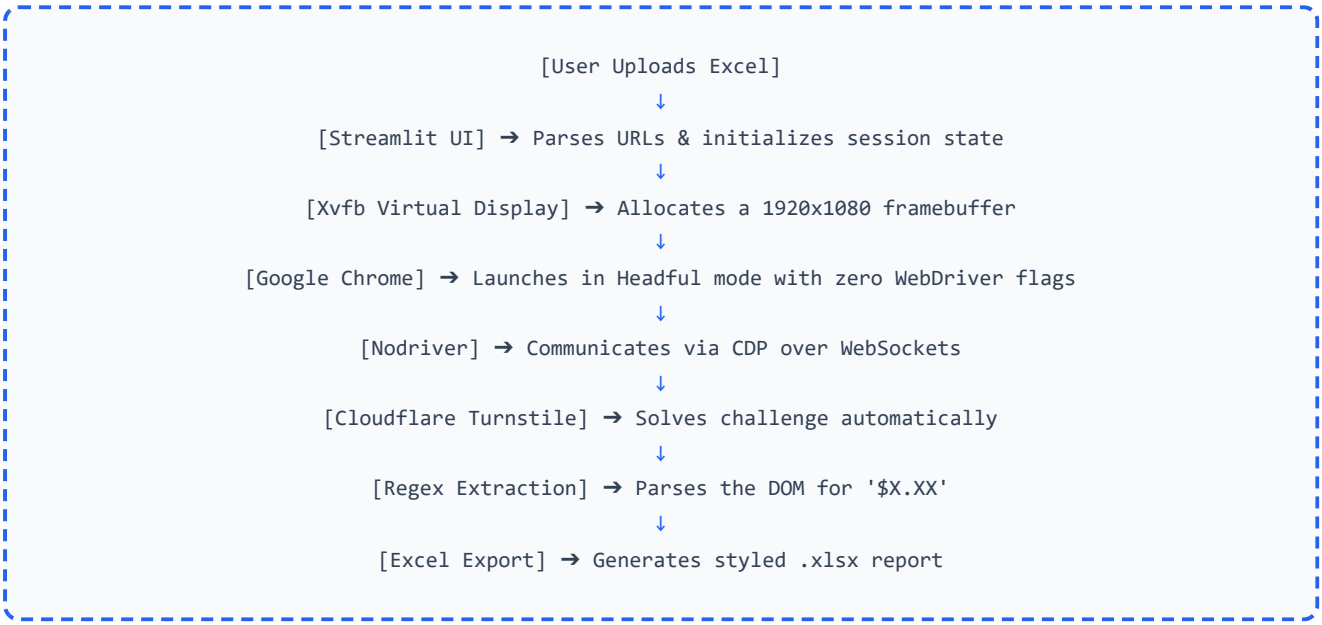
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Deploying this application requires specific configurations because Streamlit Cloud does not allow `sudo apt-get` commands and resets the environment on every boot.

Required Files:

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8. Once finished, review the metrics (Success, Balance Not Found, Errors).
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- **Regex Matching:** The regex `r"Available\s+Balance[\s\S]{0,50}?\$\s*([\d,]+\.\d{2})"` is robust, but if Tremendous changes the wording to "Current Balance", the regex must be modified.

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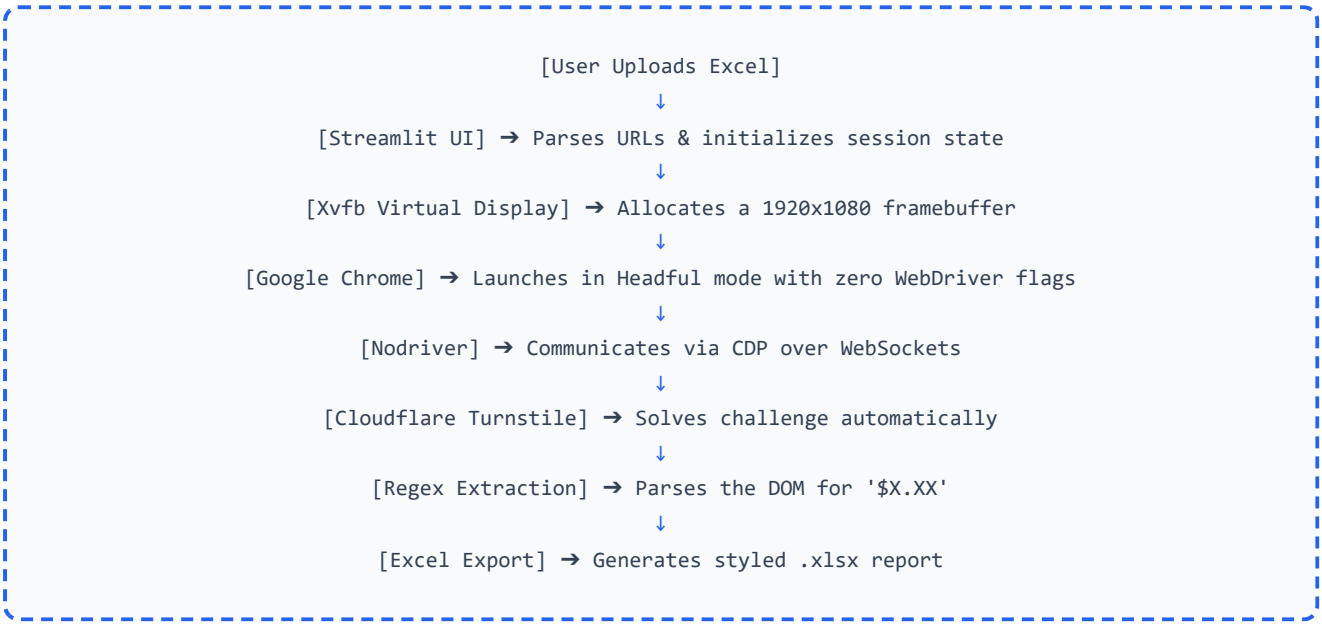
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2. System Architecture Deep Dive (Part 12)

The application is built on a multi-layered architecture designed for maximum stealth and resilience against Cloudflare's bot detection systems. Below is the component breakdown:



Each of these layers plays a critical role. If the Xvfb display fails to start, Chrome will crash. If Chrome fails to download dynamically, Nodriver will throw a `FileNotFoundError`. If Nodriver fails to connect, the Cloudflare challenge will loop infinitely.

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3. Deployment SOP: Streamlit Cloud (Part 12)

Deploying this application requires specific configurations because Streamlit Cloud does not allow `sudo apt-get` commands and resets the environment on every boot.

Required Files:

- `app.py` - The main script containing the Streamlit UI and Nodriver logic.
- `requirements.txt` - Python dependencies (`streamlit`, `nodriver`, `openpyxl`, `pandas`, `nest-asyncio`).
- `packages.txt` - Linux OS dependencies required by Chrome (e.g., `xvfb`, `libnss3`, `libatk1.0-0`, `libgbm1`, `libasound2`).

Deployment Steps:

1. Push the repository to GitHub.
2. Go to share.streamlit.io and click "New app".
3. Select the repository and branch.
4. Set the main file path to `app.py`.
5. Click "Deploy". Streamlit will install `packages.txt` first, then `requirements.txt`.

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The most critical component of this app is the Cloudflare bypass. Cloudflare uses TLS fingerprinting (JA3), Canvas hashes, and WebGL rendering to detect bots. Standard headless browsers fail all three tests.

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if "Just a moment" in page_title or "Performing security verification" in page_title:
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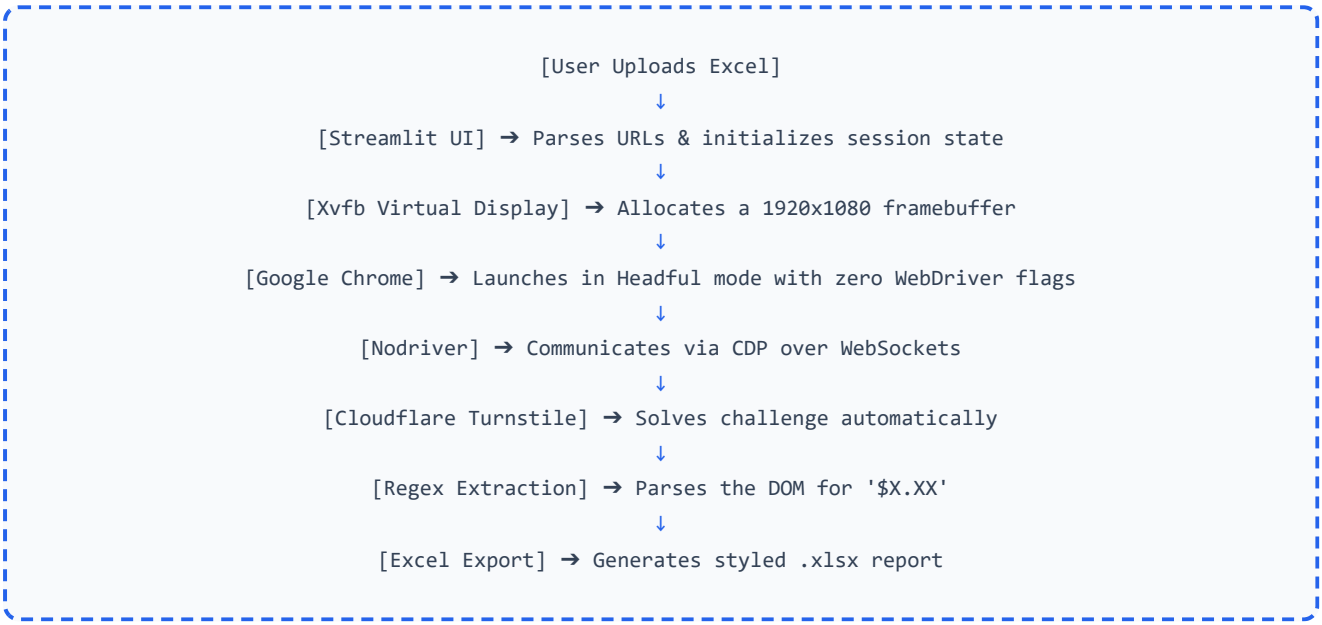
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